SYSTEM ANALYSIS DOCUMENTATION (SAD)

Project: UniGuide **Version:** 2.0 (sync with shipped state 2026-04-20) **Domain:** AI Systems & Agentic Workflow Automation (Domain 1) **Team:** Breaking Bank **Submission:** UMHackathon 2026 — Preliminary Round **Companion document:** PRD.md **Live deployment:** https://uniguide-blush.vercel.app

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Introduction

Purpose

This System Analysis Documentation (SAD) describes the technical scope, architectural decisions, data design, and operational strategy behind UniGuide — an AI-driven workflow assistant that guides Universiti Malaya students through complex multi-step administrative procedures using Z.AI's GLM as the central reasoning engine.
The document covers:

- **Architecture:** browser-based three-role app backed by serverless Next.js 15 App Router APIs, a Supabase Postgres database with `pgvector`, Supabase Storage for file uploads, Supabase Realtime for instant client updates, and Z.AI GLM as a versioned-prompt service layer.
- **Data flows:** how a student's procedure choice moves through GLM-powered turn-by-turn step emission, document upload + retrieval, coordinator briefing, and letter generation with hallucination check.
- **Model process:** end-to-end workflow for the demonstrative Scholarship & Financial Aid Application procedure — from intake through coordinator approval, including the realtime-update path back to the student.
- **Role of reference:** binding contract for the development team, reviewers, and judges to verify what is being built and why.

Background

Universiti Malaya runs hundreds of administrative procedures across faculty, postgraduate, international-student, and examination domains. Today these procedures are documented in fragmented PDFs and word-of-mouth, with three separate student-facing portals (MAYA, SiswaMail, SPeCTRUM) that don't share a single procedural status surface. Students miss deadlines they didn't know existed; staff triage submissions that are missing the same fields week after week.

Previous Version

None — this is a greenfield project built specifically for UMHackathon 2026.

Changes in Major Architectural Components (vs. Conventional University Workflow Tools)

Conventional workflow tools require an administrator to **hand-design** a workflow template before users can execute it. Templates are static; branching logic is hard-coded; routing is regex over form fields. UniGuide inverts this:

Conventional Tool	UniGuide
Admin hand-builds template upfront	**GLM emits the next step at runtime** from the indexed SOP + the student's complete history
Routing = pre-coded edges + regex over form fields	**There are no edges** — every step is a fresh `nextStep` GLM call that reads everything answered so far
Steps = static forms shown to all users identically	**Steps are emitted per-user per-turn** — type, prompt, config, citations all decided at runtime
Failure = task stuck, manual escalation	**GLM auto-fallback to fixtures** + 5-minute coordinator undo + step revise — multiple recovery paths
Coordinator decides on raw form	**Pre-digested briefing** + AI-suggest comment + previewable letter + hallucination check

Removing the GLM service layer collapses UniGuide to nothing — there is no static workflow template to fall back on. This is by deliberate design, in line with the hackathon brief.

Target Stakeholders

Stakeholder	Role	What they get
Student (Undergraduate)	Initiates a workflow for an administrative procedure.	Adaptive turn-by-turn steps with citations, real file uploads to Supabase Storage, real-time status updates, profile editor, withdraw + revise, message thread with coordinator, source-SOP viewer.
Student (Postgraduate)	Same as undergraduate, plus research-mode supervisor matching.	Same surface; procedure-specific SOP indexes provide the variation.
Student (International)	Same plus EMGS visa renewal, MoE attendance compliance.	Same surface.
Yayasan UM Scholarship Coordinator	Reviews student scholarship applications.	Inbox sorted by AI urgency, plain-English confidence labels, AI briefing per submission, AI-suggest comment, preview-and-edit letters with hallucination check, undo within 5 min, internal notes

Stakeholder	Role	What they get
		(staff-only), realtime message thread with student.
Faculty Postgraduate Committee Member	Reviews postgraduate admission applications.	Same coordinator surface.
Faculty Dean / Deputy Dean	Escalation tier; admin-level access.	All coordinator features + analytics + GLM trace audit.
Admin	Indexes SOPs, edits letter templates, sets deadlines.	Procedures library, 3-step SOP upload modal (paste / URL / **PDF**), letter template editor with sensible defaults, deadline editor, analytics dashboard, GLM reasoning-trace viewer.
Development Team	Build and maintain UniGuide.	Versioned prompts in `lib/glm/prompts/*.md`, Zod-validated GLM I/O, 14 numbered Supabase migrations, Next.js App Router with strict TypeScript.
QA Reviewer	Validate system behaviour across roles and edge cases.	Test fixtures in `tests/fixtures/glm/`, mock-mode demo seed (5 sample applications via `/api/demo/reset`).

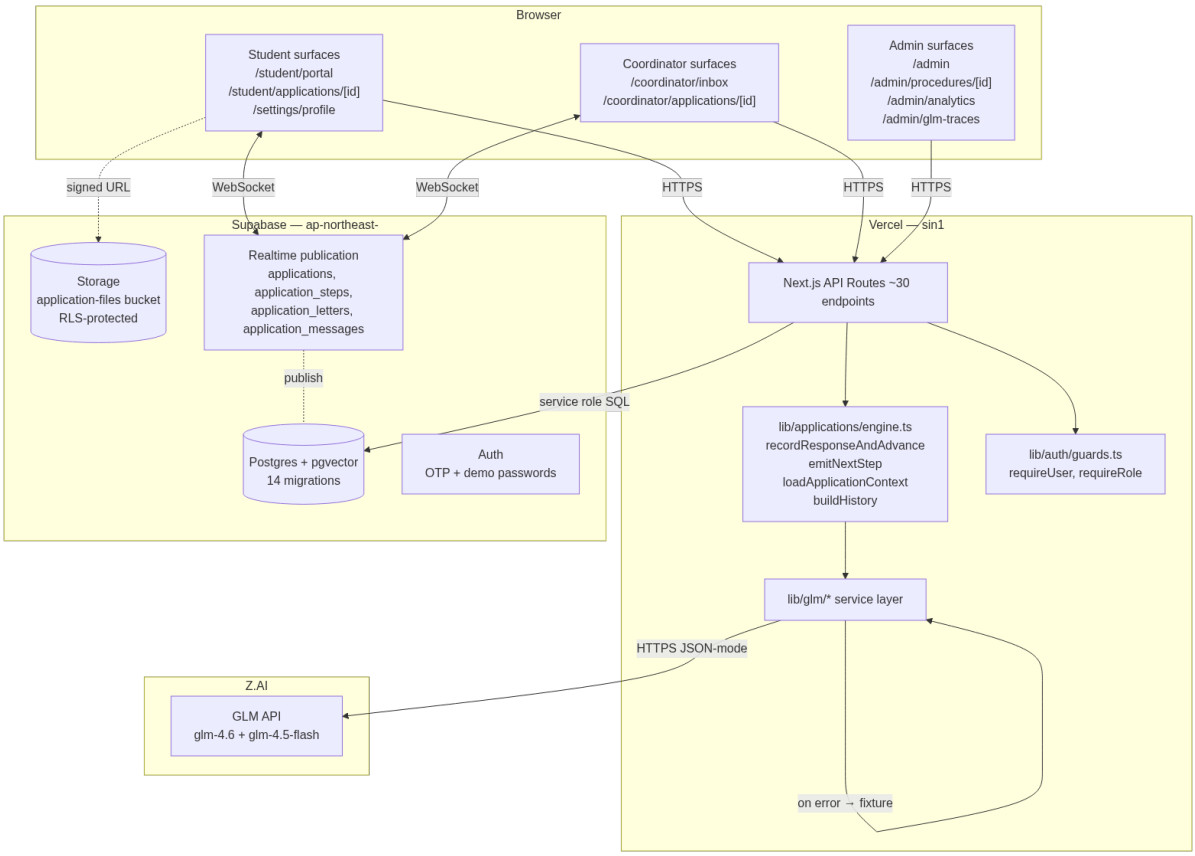
System Architecture & Design

High Level Architecture Overview

Type	Details
System	Web application (responsive on student-facing pages)
Architecture	Next.js 15 App Router (server + client components) + ~30 API route handlers + Postgres with `pgvector` + Supabase Storage + Supabase Realtime + Z.AI GLM as service layer
Hosting	Vercel (frontend + API routes), pinned to `sin1` (Singapore) for ~70ms RTT to Supabase. Supabase Cloud `ap-northeast-2` (Seoul).

Type	Details
Topology	Three role-distinct browser surfaces (Student / Coordinator / Admin) all served from a single Next.js deployment. Stateless API layer; all session state in Supabase Postgres. Realtime channels push status updates from Postgres → client without polling.

Component Diagram



LLM as Service Layer

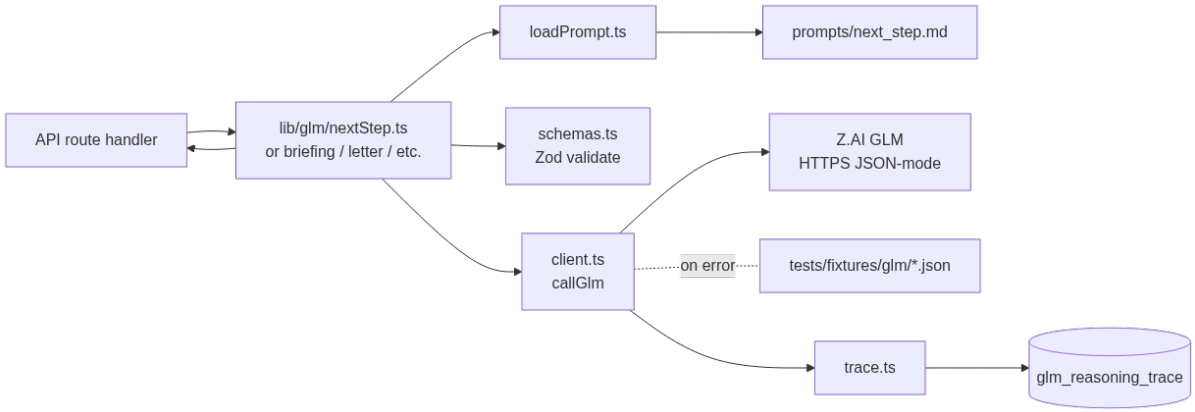
GLM is encapsulated in `lib/glm/`:

File	Purpose
`client.ts`	The single `callGlm({model, systemPrompt, userPrompt, jsonMode, mockFixture, ...})` entry point. Handles real Z.AI calls + mock-mode + auto-fallback to fixtures on error. Never imported directly outside `lib/glm/`.

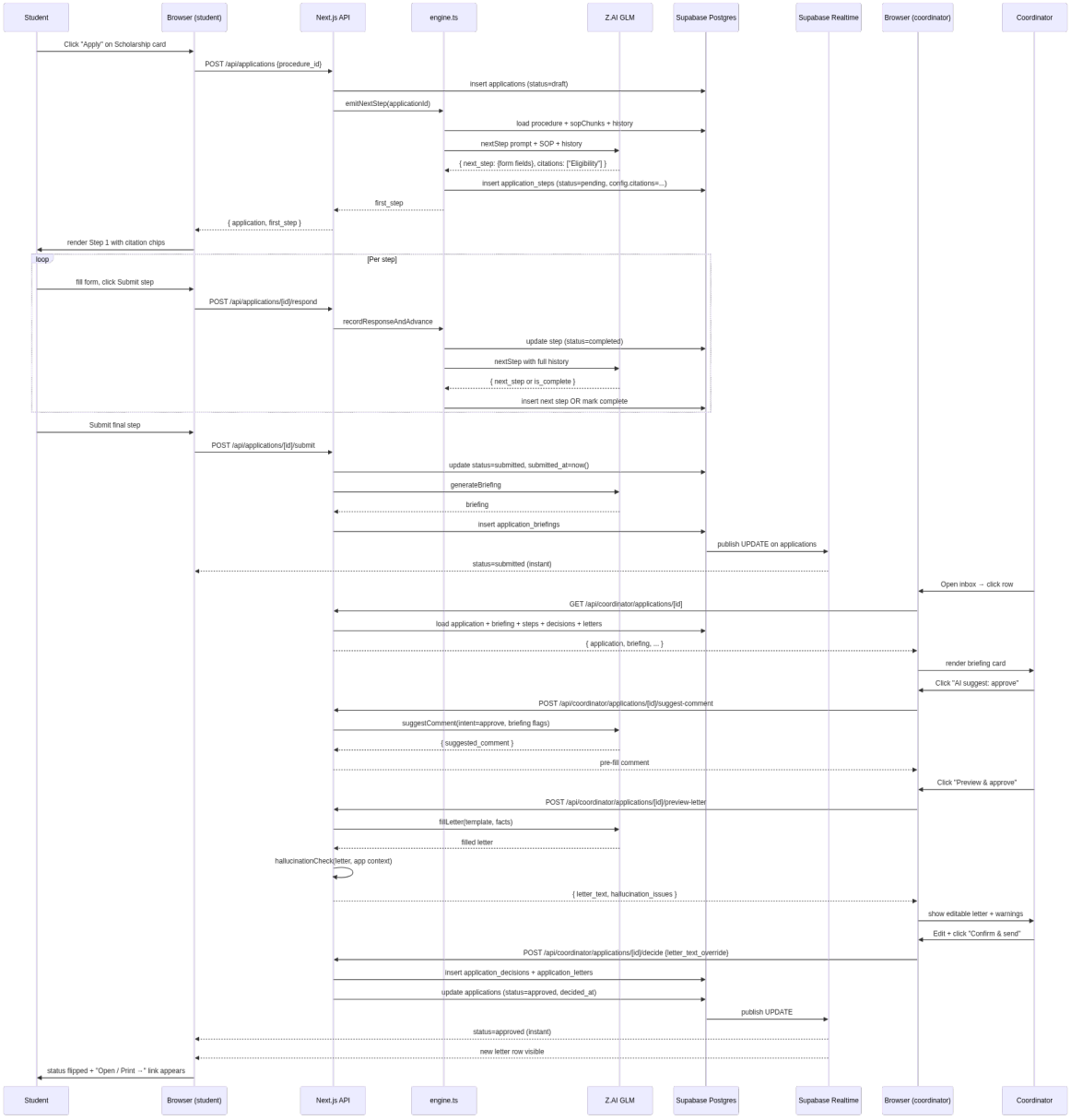
File	Purpose
`schemas.ts`	Zod schemas for every GLM input + output. `NextStepInputSchema`, `NextStepOutputSchema`, `BriefingOutputSchema`, `FillLetterOutputSchema`, `EstimateProgressOutputSchema`, `FormFieldSchema` (with file field type), `StepConfigSchema`.
`nextStep.ts`	Calls `callGlm(model: glm-4.6)` to emit the next application step. Returns `{ is_complete, next_step?, reasoning, running_summary?, citations[] }`.
`generateBriefing.ts`	Calls `callGlm(model: glm-4.6)` to produce the coordinator briefing. Returns `{ extracted_facts, flags[], recommendation, reasoning, confidence }`.
`fillLetter.ts`	Calls `callGlm(model: glm-4.6, jsonMode: true)` to fill a `{{placeholder}}` template against application context. Returns `{ filled_text, unfilled_placeholders[] }`.
`estimateProgress.ts`	Calls `callGlm(model: glm-4.5-flash)` for total step count.
(suggestComment)	Inline system prompt in `/api/coordinator/applications/[id]/suggest-comment` route. Drafts coordinator-side comment based on briefing flags + decision intent.
`loadPrompt.ts`	Loads system-prompt `.md` files from `prompts/` so prompt content is versioned in source.
`trace.ts`	After every `callGlm`, writes a row to `glm_reasoning_trace` (model_version, prompt_hash, input_summary, output, latency_ms, input_tokens, output_tokens, confidence, cache_hit).
`prompts/*.md`	The actual system prompt for each endpoint, version-controlled and hash-loggable.

Every endpoint's I/O is JSON-mode with strict Zod validation. Schema violation throws → caller returns 500. There is no free-text "freestyling" — GLM always speaks structured JSON to UniGuide.

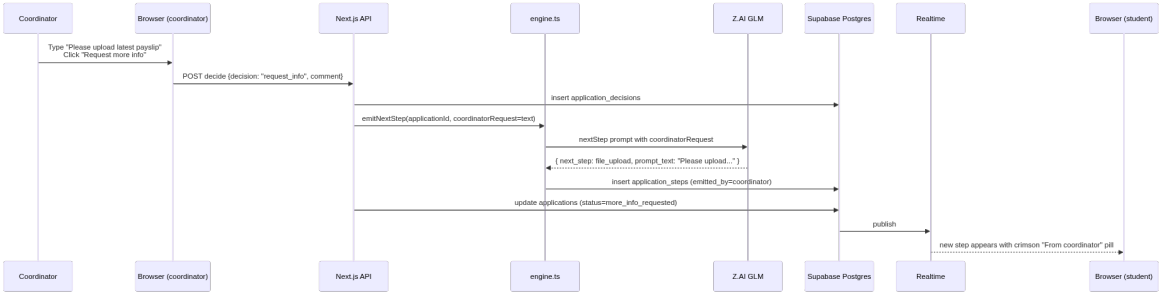
Dependency Diagram (GLM Service Layer Internals)



Sequence Diagram — Scholarship Application (Happy Path, Submit-to-Approve)



Sequence Diagram — Coordinator Requests More Info (Loop)



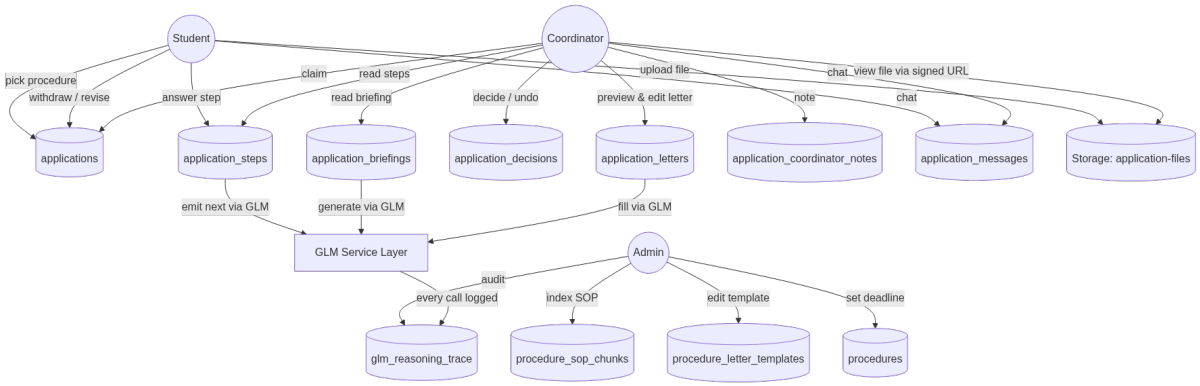
Technological Stack

Layer	Choice	Why
Frontend framework	Next.js 15 App Router (React 19)	Server components for data-heavy admin pages; client components where realtime/interactivity needed.
Language	TypeScript (strict)	Schema-driven I/O via Zod; catches GLM contract drift at compile time.
Styling	Tailwind CSS 4 + custom design tokens	Custom palette (--ink navy, --crimson, --moss, --ai-tint lavender). Component-scoped classes (`ug-card`, `ug-btn`, `ug-input`, `ug-pill`).
Icons	Lucide React	Single icon system; replaced all emoji to keep the UI professional and accessible.
Auth	Supabase Auth (OTP email + demo password)	Cookie-based JWT via `@supabase/ssr`. Three demo accounts seeded for one-click sign-in.
Database	Supabase Postgres + `pgvector`	One managed service for relational + vector. All tables RLS-protected.
Storage	Supabase Storage (`application-files` bucket)	Private bucket. RLS: INSERT requires owner folder; SELECT requires owner OR staff/admin. Signed URLs (60s) via `/api/files/sign`.
Realtime	Supabase Realtime publication	`applications`, `application_steps`, `application_letters`, `application_messages` published. Client subscribes to filtered channels.
LLM	Z.AI GLM via OpenAI-compatible SDK	`glm-4.6` for reasoning; `glm-4.5-flash` for high-volume. JSON mode + system+user prompts + max_tokens.
PDF parsing	`pdf-parse` (server-side)	Marked as `serverExternalPackages` in

Layer	Choice	Why
		`next.config.ts` to avoid webpack bundling its Node-only deps.
Hosting	Vercel (`sin1` region)	Pinned to Singapore for low latency to Supabase Seoul.

Key Data Flows

Data Flow Diagram (DFD — Level 1)



Normalized Database Schema (3NF — ERD)

14 tables across 13 migrations:

```
users (id PK, email UNIQUE, role CHECK in admin/student/staff, created_at)

student_profiles (user_id PK→users.id, full_name, faculty, programme,
    year CHECK 1-8, cgpa CHECK 0-4, citizenship default MY, matric_no UNIQUE)

staff_profiles (user_id PK→users.id, full_name, faculty,
    staff_role CHECK in admin/coordinator/dean/dvc/ips_officer)

procedures (id PK text, name, description, source_url, faculty_scope,
    indexed_at, deadline_date, deadline_label)

procedure_sop_chunks (id PK uuid, procedure_id→procedures.id, chunk_order,
    section, content, source_url, embedding vector(1536), indexed_at)

procedure_letter_templates (id PK uuid, procedure_id→procedures.id,
    template_type CHECK in acceptance/rejection/request_info/custom,
    name, template_text, detected_placeholders text[],
    created_by→users.id, created_at, updated_at,
    UNIQUE(procedure_id, template_type, name))
```

```
applications (id PK uuid, user_id→users.id, procedure_id→procedures.id,
    status CHECK in draft/submitted/under_review/more_info_requested/
        approved/rejected/withdrawn,
    progress_current_step, progress_estimated_total,
    student_summary, ai_recommendation, ai_confidence,
    assigned_to→auth.users(id), assigned_at,
    created_at, submitted_at, decided_at, updated_at)

application_steps (id PK uuid, application_id→applications.id, ordinal,
    type CHECK in form/file_upload/text/select/multiselect/info/
        final_submit/coordinator_message,
    prompt_text, config jsonb, emitted_by CHECK in ai/coordinator,
    emitted_by_user_id→users.id, status CHECK in pending/completed/skipped,
    response_data jsonb, created_at, completed_at)

application_briefings (id PK uuid, application_id→applications.id,
    extracted_facts jsonb, flags jsonb default '[]',
    recommendation CHECK, reasoning,
    status CHECK in pending/resolved, created_at)

application_decisions (id PK uuid, application_id→applications.id,
    briefing_id→application_briefings.id, decided_by→users.id,
    decision CHECK in approve/reject/request_info/withdrawn,
    comment, decided_at)

application_letters (id PK uuid, application_id→applications.id,
    template_id→procedure_letter_templates.id,
    letter_type CHECK, generated_text, pdf_storage_path,
    delivered_to_student_at, delivered_via_email default false, created_at)

application_coordinator_notes (id PK uuid,
    application_id→applications.id, author_id→auth.users(id),
    body CHECK length 1-4000, created_at, updated_at)
    [RLS: staff/admin read+insert, author delete-own]

application_messages (id PK uuid, application_id→applications.id,
    author_id→auth.users(id),
    author_role CHECK in student/coordinator,
    body CHECK length 1-4000, created_at)
    [RLS: owner+staff read; insert by role with author_role match]
    [Realtime publication: yes]

glm_reasoning_trace (id PK uuid, workflow_id, endpoint CHECK,
    model_version, prompt_hash, input_summary jsonb, output jsonb,
    confidence, citations text[], citation_verified default true,
    input_tokens, output_tokens, latency_ms, cache_hit default false,
    retry_count default 0, called_at)

attachments (legacy v1 table — superseded by Storage bucket + step.response_data)
```

Storage:

- `application-files` bucket (private, 10 MB limit, MIME-restricted to PDF/JPG/PNG/WEBP/DOC/DOCX)

- Path: `{user_id}/{application_id}/{step_id}-{timestamp}-{safeName}`
- RLS:
- INSERT: `(storage.foldername(name))[1] = auth.uid()::text`
- SELECT: same OR staff/admin
- DELETE: owner only

Realtime publication (`supabase_realtime`): `applications`, `application_steps`, `application_letters`, `application_messages`.

API Surface (~30 endpoints)

Method	Path	Auth	Purpose
GET	`/api/procedures`	any user	List procedures for stude
GET	`/api/procedures/[id]/sop`	any user	SOP chunks for the SOP v
GET, POST	`/api/applications`	student	List my apps / create new
GET	`/api/applications/[id]`	owner+staff	Full application
POST	`/api/applications/[id]/respond`	owner	Submit step response →
POST	`/api/applications/[id]/submit`	owner	Final submit → trigger br
POST	`/api/applications/[id]/withdraw`	owner	Cancel an in-flight app
POST	`/api/applications/[id]/revise/[stepId]`	owner	Reset a completed step + steps
GET, POST	`/api/applications/[id]/messages`	owner+staff	Message thread
GET	`/api/coordinator/inbox`	staff/admin	Inbox queue with briefing
GET	`/api/coordinator/applications/[id]`	staff/admin	Detail view with briefing/steps/decisions/
POST	`/api/coordinator/applications/[id]/decide`	staff/admin	Approve / reject / reques optional `letter_text_`
POST	`/api/coordinator/applications/[id]/preview-letter`	staff/admin	Generate letter without c run hallucination check
POST	`/api/coordinator/applications/[id]/suggest-comment`	staff/admin	AI-draft a coordinator co
POST	`/api/coordinator/applications/[id]/undo`	staff/admin	Revert a decision within 5
POST, DELETE	`/api/coordinator/applications/[id]/claim`	staff/admin	Claim / release an applica
GET, POST	`/api/coordinator/applications/[id]/notes`	staff/admin	Internal notes

Method	Path	Auth	Purpose
DELETE	`/api/coordinator/applications/[id]/notes/[noteId]`	author	Delete own note
GET, POST	`/api/admin/procedures`	admin	List / create procedures
PATCH	`/api/admin/procedures/[id]`	admin	Update deadline, descrip faculty_scope
POST	`/api/admin/procedures/[id]/sop`	admin	Upload SOP text → chunl
POST	`/api/admin/procedures/parse-pdf`	admin	Extract text from a PDF (p
GET, POST	`/api/admin/procedures/[id]/letter-templates`	admin	List / upsert templates
DELETE	`/api/admin/procedures/[id]/letter-templates/[templateId]`	admin	Delete template
GET	`/api/files/sign?path=…`	owner+staff	60s signed URL for an upl
GET	`/api/notifications`	any user	Last-14-days events feed
GET, POST	`/api/profile`	any user	Read / upsert profile
POST	`/api/profile/bootstrap`	any user	Returns whether onboard
POST	`/api/demo/reset`	any user	Wipe + reseed demo stuc sample apps

Functional Requirements & Scope

Minimum Viable Product (MVP)

Live demo procedure: Scholarship & Financial Aid (Yayasan UM) — implemented end-to-end.
All 28 functionalities (F1–F28 in PRD §4.2) are wired and demoable. The demo seeds 5 sample applications (mid-flow draft, high-conf approve, low-conf+flagged, approved, rejected) so coordinator inbox + analytics + GLM traces all populate immediately on `/api/demo/reset`.
Six procedures listed in the catalogue; only Scholarship is Live (others get "Coming soon" badges so the catalogue communicates scope without misleading).

Non-Functional Requirements (NFRs)

NFR	Target	How achieved
Latency p95 — `nextStep`	< 4s	Model: glm-4.6, JSON mode, max_tokens 2000. Vercel `sin1` ↔ Supabase Seoul ~70ms.

NFR	Target	How achieved
Latency p95 — `suggestComment`	< 1.5s	Model: glm-4.5-flash, max_tokens 400.
Realtime push p95	< 1s	Supabase Realtime publication; client-side subscription with filter.
File upload p95	< 3s for 5 MB	Direct browser-to-Storage via Supabase JS client; no Next.js relay.
Auth p95	< 500ms	Cookie-based JWT verified via `@supabase/ssr`.
Security — file privacy	Owner + staff only	Bucket RLS + signed URLs. Verified by demo cross-account test.
Audit — every GLM call	100%	`trace.ts` writes `glm_reasoning_trace` for every `callGlm` invocation. Visible at `/admin/glm-traces`.
Resilience — GLM outage	Demo continues	Auto-fallback to fixtures with logged error; `model: …-fallback` marker in trace.

Out of Scope / Future Enhancements

- Email delivery via SMTP (currently in-app letter delivery)
- Server-side PDF generation (currently browser print)
- OCR for image-only PDFs
- Bahasa Melayu UI
- Voice intake
- Real MAYA / SiswaMail / EMGS API integrations
- Payment processing
- Multi-tenant isolation
- Native mobile apps
- Push notifications
- Coordinator-side bulk reject

Monitor, Evaluation, Assumptions & Dependencies

Technical Evaluation

Rollout Strategy

- **Mock mode default** (`GLM_MOCK_MODE=true`) on shared deploy + demo banner. Real Z.AI mode is one env var flip on Vercel (`GLM_MOCK_MODE=false` + valid `ZAI_API_KEY`).
- **Auto-fallback on real-mode error** — even after flipping to real, if a call errors, the fixture saves the demo. Logged loudly to `console.error` so production monitoring catches it.

Emergency Rollback

- Each round of features was committed + pushed independently (~12 commits across 7 rounds). Vercel keeps prior deployments — one-click rollback via dashboard.
- Database migrations are additive (no destructive ALTERs); rollback typically only needs Vercel rollback, not migration reversal.

Priority Matrix (what gets fixed first if demo breaks)

Priority	Surface	Why
P0	Student demo flow (`/login` → `/student/portal` → application → submit)	Core narrative for judges
P0	Coordinator demo flow (`/login` → `/coordinator/inbox` → detail → preview & approve)	Other half of core narrative
P1	`/admin/glm-traces`	Transparency story; judges may ask
P1	Realtime updates	Wow factor; if broken, flag manually
P2	Admin SOP upload	Less likely demoed live

Monitoring

- Vercel deployment logs + runtime logs
- `glm_reasoning_trace` for every model call (latency, tokens, errors)
- Supabase dashboard for DB + Storage metrics
- `console.error` from auto-fallback path captured in Vercel logs

Assumptions

- UM SOPs change infrequently enough that re-indexing once a semester is sufficient.
- Coordinators are willing to read briefings + edit GLM-drafted letters (vs. full automation, which we deliberately don't do).
- Students are willing to accept turn-by-turn step emission as a flow (vs. seeing the whole form upfront — we believe step-at-a-time reduces overwhelm).
- Z.AI GLM availability is sufficient for production scale; rate-limit + cost monitoring is in roadmap.

External Dependencies

- **Z.AI GLM API** — mandatory per hackathon rules. Mock-mode + auto-fallback hedges against outages.
- **Supabase Cloud (Postgres + Storage + Auth + Realtime)** — managed hosting. Single-region (Seoul) for now.
- **Vercel** — Next.js hosting. Pinned to `sin1`.
- **`pdf-parse`** — server-only PDF text extraction. Marked `serverExternalPackages` in `next.config.ts`.

Project Management & Team Contributions

Project Timeline

- **Week 1 (Apr 14–18):** v1 architecture (workflow templates + ReactFlow canvas), 3 documents, initial Supabase schema, login + onboarding.
- **Week 2 (Apr 18–20):** v2 pivot to step-by-step AI emission. 14 migrations. 7 rounds of feature work covering all surfaces.
- **Apr 20:** Documents synced to actual shipped state (this revision).
- **Apr 20–25:** Demo polish, rehearsals, video record.
- **Sat Apr 26 07:59:** Hackathon submission deadline.

Team Members & Roles

- **Jeanette Tan En Jie** — Frontend, design system, copy
- **Teo Zen Ben** — Backend, GLM integration, infrastructure (Vercel + Supabase)
- **Nyow An Qi** — Procedure research, SOP indexing, fixture authoring
- **Thevesh A/L Chandran** — Coordinator UX, briefing prompt engineering, demo seed

Recommendations

- **For real production use:** wire SMTP for letter delivery; add OCR for scanned PDFs; harden bucket RLS with per-procedure scoping; cost-monitor GLM with daily quota alerts.
- **For other Malaysian universities:** the engine is procedure-agnostic — swap the SOP corpus + letter templates and the same UniGuide app guides USM / UKM / UPM / UTM students.
- **For other Malaysian government procedures:** SSM / LHDN / EPF / Halal cert all have the same shape — UniGuide's pattern (SOP → AI emits steps → coordinator decides) generalises.

End of SAD v2.0 — synced with shipped state as of 2026-04-20